Functional Requirements

Detailed project document · Prepared for project completion and review

Project reference information

Item	Known project detail
Project	Health360 Clinical & Healthcare Risk Analysis
Dataset size	5,000 patient/person records
Dataset fields	17 columns; exact field names and definitions must be copied from the source dataset's header row/data dictionary
Analytics tool	Tableau
Visual assets	9 worksheets/visualizations and a dashboard; a Tableau story for Patient Risk & Lifestyle Insights was also part of the project workflow
Repository	https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git
Website	https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/
Scope caution	This is an educational data-visualization project. It is not a clinically validated diagnostic, prognostic, or treatment system.

FR-01: data inventory

The project shall record source file name, publisher, retrieval date, license/terms, record count, field count, and the exact 17 column names. Acceptance test: compare the documented inventory to the actual source file and record any mismatch.

FR-02: data validation

The project shall check data types, missingness, duplicates, inconsistent category spellings, invalid values, and unit consistency. Each correction or exclusion must be recorded. Acceptance test: produce a before/after summary and verify that no change is unexplained.

FR-03: metric definitions

Each measure used in Tableau shall have a documented definition, aggregation, denominator where applicable, unit, and handling of null values. Percentages must specify the population included in the denominator. Acceptance test: manually recalculate sample cases.

FR-04: worksheets

The workbook shall contain nine purposeful worksheets/visualizations as stated in the project workflow. For each worksheet, document the analytical question, dimensions, measures, filters, encoding, and expected interpretation. The exact worksheet titles and measures must match the actual workbook; do not invent titles in the final report.

FR-05: dashboard

The dashboard shall combine selected worksheets, provide a clear reading order, use consistent formatting, and expose only meaningful filters/actions. Acceptance test: test each filter with multiple selections, verify cross-filtering if enabled, and test reset behavior.

FR-06: story

The story shall sequence views or annotations so that each point adds context. Every claim must be supported by the displayed data. Acceptance test: review story point order and confirm that each point can be understood without unsupported assumptions.

FR-07: publication and website

The published dashboard/story links and website shall open for the intended audience. If public sharing is used, confirm the data is approved for public release. Acceptance test: open links in a signed-out/private browser session and test on desktop and mobile.

FR-08: documentation

The project shall include a final report, data dictionary, cleaning log, test record, references, demonstration script, and a handoff inventory. Acceptance test: check that the filenames, workbook version, and URLs agree across documents.

Completion and review record

Prepared by: _____ Date: _____

Reviewed by: _____ Status: Draft / Reviewed / Approved

Document-control note: Update this file when the dataset, workbook, published views, or website changes. Record version/date and keep claims tied to evidence. This detailed document uses known project context and explicitly flags details that must be verified from the actual source files.